

# Test Results

Evidence from the automated role, permission and end-to-end business flow test suites.

|          |                                       |
|----------|---------------------------------------|
| Audience | Management, project team              |
| Suites   | Role smoke test, full end-to-end flow |
| Result   | 152 of 152 checks passed              |
| Last run | 3 July 2026                           |

# Tefoma Procurement — Smoke & E2E Test Results

**Product:** Tefoma Procurement **Last run:** 3 July 2026 **Environment:** Localhost ( `http://localhost:3001` API, MongoDB seeded via `npm run seed:all -w api` ) **Status:** All automated checks **passed**

## Executive summary

| Suite                   | Command                               | Result                                  |
|-------------------------|---------------------------------------|-----------------------------------------|
| Role smoke test         | <code>npm run smoke -w api</code>     | 103 / 103 PASS                          |
| Full E2E flow           | <code>npm run e2e:local -w api</code> | 49 / 49 PASS (incl. supplier analytics) |
| Client production build | <code>npm run build -w client</code>  | PASS                                    |

These tests exercise the **API layer** (authentication, role permissions, and full procurement workflow). Browser UI clicks are **not** automated — use `CLIENT_TESTING_GUIDE.md` for manual UI walkthrough.

## Prerequisites (localhost)

1. MongoDB running locally.
2. Copy env files (if not already present):
- `api/.env` from `api/.env.example` — set `MONGODB_URI`, `JWT_SECRET`, and `OTP_EXPOSE_IN_RESPONSE=true` for automated OTP login.
  - `client/.env` — `VITE_API_URL=http://localhost:3001/api`
1. Seed test data:
- ```
`` bash npm run seed:all -w api ``
```
1. Start API (required before tests):
- ```
`` bash npm run dev -w api ``
```
- Seeded password (all staff & supplier test accounts): `Admin@123`

## How to run

```
# Role + endpoint smoke test (default: localhost:3001)
npm run smoke -w api

# Against production (only if accounts exist there)
node api/src/scripts/smoke-test.mjs https://fosssil-procure-api.vercel.app

# Full business-flow E2E (supplier onboarding → payment)
npm run e2e:local -w api
```

## Test accounts used

| Email                   | Role                          |
|-------------------------|-------------------------------|
| admin@fossilzim.com     | Admin                         |
| macdonald@fossilzim.com | Procurement officer           |
| jb@fossilzim.com        | Department head (Procurement) |
| mac@fossilzim.com       | Department head (ICT)         |
| james@fossilzim.com     | End user (ICT)                |
| paul@fossilzim.com      | Finance                       |
| тино@fossilzim.com      | COO                           |
| alfred@fossilzim.com    | Stores officer                |
| ict.hw@techzone.co.zw   | Supplier (TechZone Hardware)  |
| ict.sw@codebridge.co.zw | Supplier (used in E2E quotes) |
| ict.maint@profix.co.zw  | Supplier (used in E2E quotes) |

## Suite 1 — Role smoke test (103 checks)

Script: `api/src/scripts/smoke-test.mjs`

### Public / unauthenticated (6 checks) — ALL PASS

| Check                                                                                   | Result |
|-----------------------------------------------------------------------------------------|--------|
| <code>GET /health</code>                                                                | PASS   |
| <code>GET /api/</code>                                                                  | PASS   |
| <code>POST /auth/login</code> invalid credentials → 401                                 | PASS   |
| <code>POST /auth/login</code> missing fields → 400                                      | PASS   |
| <code>GET /auth/me</code> without token → 401                                           | PASS   |
| <code>OPTIONS /auth/login</code> CORS for <code>https://fossilprocure.vercel.app</code> | PASS   |

### Per-role login + OTP + endpoints — ALL PASS

Each role: login → OTP verify → `GET /auth/me` → shared endpoints → role-specific GETs.

| Role                | Login | Shared endpoints                | Role endpoints                                                                                 | Total |
|---------------------|-------|---------------------------------|------------------------------------------------------------------------------------------------|-------|
| Admin               | PASS  | dashboard, notifications, sites | users, departments, sites, audit-logs                                                          | 10    |
| Procurement officer | PASS  | ✓                               | suppliers, requisitions, rfqs, quotations, POs, evaluations/due, evaluations, supplier-reports | 14    |

| Role                    | Login | Shared endpoints | Role endpoints                                                                               | Total |
|-------------------------|-------|------------------|----------------------------------------------------------------------------------------------|-------|
| Dept head (Procurement) | PASS  | ✓                | requisitions, pending PO approvals, store requisitions, catalog search                       | 10    |
| Dept head (ICT)         | PASS  | ✓                | requisitions, pending PO approvals, store requisitions                                       | 9     |
| End user                | PASS  | ✓                | requisitions, catalog search, cancellation-meta                                              | 9     |
| Finance                 | PASS  | ✓                | pending approvals, POs, invoices, payments, budgets                                          | 11    |
| COO                     | PASS  | ✓                | pending approvals                                                                            | 7     |
| Stores officer          | PASS  | ✓                | inventory, deliveries, pending deliveries, PR queue, movements, store reqs, transfers, sites | 14    |
| Supplier                | PASS  | ✓                | profile, rfqs, quotations, POs, deliveries, invoices                                         | 12    |

### Security — PASS

| Check                           | Result |
|---------------------------------|--------|
| End user GET /admin/users → 403 | PASS   |

### Suite 2 — Full E2E flow (49 checks)

Script: `api/src/scripts/e2e-localhost.mjs`

Covers the complete procurement lifecycle plus supplier onboarding, **supplier analytics**, and ancillary features.

### Phase 1 — Supplier onboarding (5 checks) — ALL PASS

| Step                                                     | Result |
|----------------------------------------------------------|--------|
| Supplier self-registration ( POST /auth/register )       | PASS   |
| New supplier views profile                               | PASS   |
| Procurement sees pending supplier                        | PASS   |
| Procurement activates supplier (KYS override + activate) | PASS   |
| Supplier updates profile                                 | PASS   |

### Phase 2 — Requisition → payment (26 checks) — ALL PASS

| Step                       | Role     | Result |
|----------------------------|----------|--------|
| Create requisition (draft) | End user | PASS   |
| Submit requisition         | End user | PASS   |
| Approve requisition        | ICT HOD  | PASS   |
| Forward to procurement     | Stores   | PASS   |

| Step                                       | Role             | Result                 |
|--------------------------------------------|------------------|------------------------|
| Accept requisition                         | Procurement      | PASS                   |
| Create & publish RFQ (3 invited suppliers) | Procurement      | PASS                   |
| Submit quotation (×3 suppliers)            | Suppliers        | PASS                   |
| Close RFQ                                  | Procurement      | PASS                   |
| Select winning quotation + justification   | ICT HOD          | PASS                   |
| Authorize quotation                        | Procurement      | PASS                   |
| Accept quotation                           | Procurement      | PASS                   |
| Create purchase order                      | Procurement      | PASS                   |
| Submit PO for approval                     | Procurement      | PASS                   |
| Approve PO                                 | HOD              | PASS                   |
| Approve PO                                 | Finance          | PASS                   |
| Approve PO (≥ USD 5,000 threshold)         | COO              | PASS (PO total ~7,130) |
| Acknowledge PO                             | Winning supplier | PASS                   |
| Receive goods (GRV)                        | Stores           | PASS                   |
| Delivery accepted (auto on receive)        | Stores           | PASS                   |
| Submit invoice                             | Supplier         | PASS                   |
| Approve invoice                            | Finance          | PASS                   |
| Record payment                             | Finance          | PASS                   |

### Phase 3 — Additional scenarios (8 checks) — ALL PASS

| Step                                         | Result |
|----------------------------------------------|--------|
| Record supplier evaluation (7 criteria, 1–5) | PASS   |
| Evaluation overallScore computed             | PASS   |
| Finance views budgets                        | PASS   |
| Admin views audit logs                       | PASS   |
| End user has notifications                   | PASS   |
| End user cancels own draft requisition       | PASS   |
| Procurement dashboard stats                  | PASS   |

### Phase 3b — Supplier analytics (Performance, Compliance, Evaluations) — ALL PASS

These three UI pages under **Suppliers** → **Analytics** share the same backend report API plus evaluation endpoints.

| UI page     | Route                                | API(s)                                                                                               | What was tested                                                                      | Result |
|-------------|--------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|
| Performance | /app/suppliers/analytics/performance | GET /procurement/supplier-reports , GET /procurement/evaluations/due                                 | Summary score bands; registry overallScore after evaluation; ranked suppliers        | PASS   |
| Compliance  | /app/suppliers/analytics/compliance  | GET /procurement/supplier-reports                                                                    | Summary kysVerified / kysPending ; registry kysPercent , kysComplete , documentCount | PASS   |
| Evaluations | /app/suppliers/evaluations           | GET /procurement/evaluations/due , GET /procurement/evaluations , POST .../suppliers/:id/evaluations | Due list; create evaluation; list history; per-supplier evaluations                  | PASS   |

#### Metrics validated (from getSupplierReports )

| Metric                                                | Used on                                | Source                                  |
|-------------------------------------------------------|----------------------------------------|-----------------------------------------|
| overallScore (1–5)                                    | Performance ranking, supplier header   | Latest SupplierEvaluation.overallScore  |
| scoreBands (excellent / good / watch / low / unrated) | Performance summary cards              | Aggregated from scores                  |
| kysVerified , kysPending                              | Compliance summary                     | SupplierProfile.kysComplete             |
| kysPercent                                            | Compliance table                       | computeKysCompletion() on KYS checklist |
| documentCount                                         | Compliance table                       | complianceDocuments.length              |
| poCount , poSpend                                     | Reports registry (also on Reports tab) | Completed PO totals                     |
| nextEvaluationDue                                     | Performance / Evaluations due tab      | SupplierProfile.nextEvaluationDue       |

#### Evaluation criteria (7 scores, 1–5 each)

Credit terms, contractual agreements, market reputation, pricing, delivery efficiency, ease in dealings, consistent quality — averaged into **overall score out of 5** (not a percentage).

Evaluations save **immediately** as **approved** (no HOD/SEC approval chain on new records).

#### Automated check list (Phase 3b)

| Check                                         | Result |
|-----------------------------------------------|--------|
| GET /procurement/supplier-reports returns 200 | PASS   |

| Check                                                           | Result |
|-----------------------------------------------------------------|--------|
| Summary <code>totalSuppliers &gt; 0</code>                      | PASS   |
| Summary includes KYS and score band counts                      | PASS   |
| Registry rows include <code>kysPercent</code>                   | PASS   |
| Evaluated supplier shows <code>overallScore ≥ 3.5</code>        | PASS   |
| GET <code>/procurement/evaluations/due</code> returns due list  | PASS   |
| GET <code>/procurement/evaluations</code> lists E2E evaluation  | PASS   |
| GET <code>/procurement/suppliers/:id/evaluations</code> history | PASS   |

### Manual UI only (not automated)

- Performance / Compliance **pagination** and row click-through to supplier profile
- Evaluations **Create evaluation** modal (sliders, recommendation dropdown)
- Compliance **filter by status** on Suppliers list
- KYS document upload impact on `kysPercent` (E2E uses KYS override for onboarding instead)

## What is NOT covered by automation

The following still require **manual UI testing** in the browser ( `npm run dev -w client` → <http://localhost:5173>):

- All 57 client routes (forms, tabs, modals, validation messages)
- KYS document **file uploads** (API override used in E2E instead)
- Store requisition / stock transfer UI flows (separate from main PR→PO path)
- Quotation waiver UI (E2E uses 3 real quotes instead)
- Email delivery (OTP appears in API console when `OTP_EXPOSE_IN_RESPONSE=true` )
- Production environment (production DB may not have all seeded accounts)

See `CLIENT_TESTING_GUIDE.md` and `TESTING_GUIDE.md` for manual test steps.

## Production note

A smoke run against <https://fossil-procure-api.vercel.app> on the same date showed:

- Public endpoints and **admin** login: **PASS**
- Other seeded accounts: **FAIL** (401 — accounts not present or different passwords in production DB)

Run production smoke only after seeding or creating matching users in the live database.

## Related documents

| Document                | Purpose                              |
|-------------------------|--------------------------------------|
| SYSTEM_OVERVIEW.md      | What the system does (business view) |
| TESTING_GUIDE.md        | Step-by-step manual E2E for testers  |
| CLIENT_TESTING_GUIDE.md | Live-site UAT guide                  |
| PROCEDURE_COMPLIANCE.md | Policy vs implementation matrix      |

*Re-run tests after major changes and update the "Last run" date and summary tables above.*